Subspaces

Definition

Let H be a nonempty collection of vectors in $\mathbb{R}^n$. Then H is a subspace if whenever a and b are scalars and $\vec{u}$ and $\vec{v}$ are vectors in H, $a\vec{u} + b\vec{v}$ is also in H.

Theorem

Let H be a nonempty collection of vectors in $\mathbb{R}^n$. Then H is a subspace of $\mathbb{R}^n$ if and only if there exist vectors $\{\vec{u}_1, \ldots, \vec{u}_k\}$ in H such that

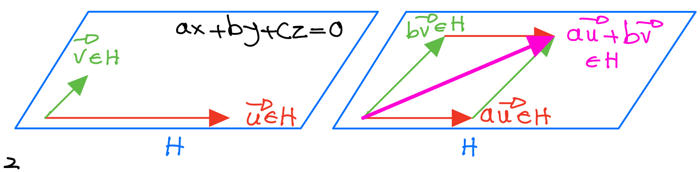
$$H = span\{\vec{u}_1, \ldots, \vec{u}_k\}$$

Example: Among the collection of vectors described below, which one is a subspace of $\mathbb{R}^n$?

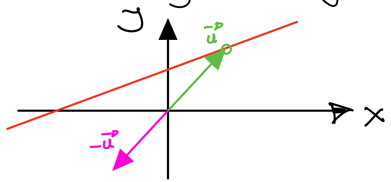
Ⓐ A plane through the origin in $\mathbb{R}^3$.

(I) This subset of $\mathbb{R}^3$ contains the zero vector.

(II) If two vectors are on this plane, so is any linear comb. of them.



Ⓑ A line not containing the origin in $\mathbb{R}^2$.



Note that, for example, $\vec{u}$ is in the subset of vectors drawn in red, however, $-\vec{u}$ is not. Hence this subset of vectors in $\mathbb{R}^2$ is NOT a subspace of $\mathbb{R}^2$.

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True or False?

Any subspace H of $\mathbb{R}^n$ contains the zero vector.

This is true. Note that since a subspace is nonempty, it has at least one element $\vec{u}$. But then $0\vec{u} = \vec{0}$ is in H since H is closed under scalar multiplication.

Is the set $T = \left\{ \begin{bmatrix} x-1 \\ x+1 \end{bmatrix}, x \in \mathbb{R} \right\}$ a subspace of $\mathbb{R}^2$?

No, clearly $\vec{0} \notin T$, since $\begin{bmatrix} x-1 \\ x+1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ is an inconsistent system.

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Is the set $T = \left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} t : t \in \mathbb{R} \right\}$ a subspace of $\mathbb{R}^3$?

Yes. (I) Choose $t=0$, to see that $\vec{0} \in T$.

(II) For any $\vec{u}$ and $\vec{v}$ in T, $a\vec{u} + b\vec{v}$ is also in T:

If $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} t_1$ and $\vec{v} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} t_2$, Then $a\vec{u} + b\vec{v} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} at_1 + \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} bt_2$

$$= \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} (at_1 + bt_2)$$

$$= \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} t_3 \text{, where } t_3 \in \mathbb{R}.$$

Examples of subspaces of $\mathbb{R}^3$

(I) $H_1 = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : x = y = 2z \right\}$

Note that this is a line that passes through the origin and can be thought of as $H_1 = \text{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$.

(II) $H_2 = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : x - y = 0 \right\}$

Note that this is a plane that passes through the origin and can be thought of as $H_2 = \text{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$.

(III) $H_3 = \mathbb{R}^3 = \text{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

$$= \text{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \right\}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \left(\frac{x+y}{2}\right) \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \left(\frac{x-y}{2}\right) \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Theorem (Subspaces are spans and spans are subspaces)

Let H be a nonempty collection of vectors in $\mathbb{R}^n$. Then H is a subspace of $\mathbb{R}^n$ if and only if there exist vectors $\{\vec{u}_1, \ldots, \vec{u}_k\}$ in H such that

$$H = span\{\vec{u}_1, \ldots, \vec{u}_k\}$$

Proof: We only show that if H is a subspace, then $H = \text{Span}\{\vec{u}_1, \ldots, \vec{u}_k\}$. Pick a vector $\vec{u}_1 \in H$. If $H = \text{Span}\{\vec{u}_1\}$, then we are done. If $H \neq \text{Span}\{\vec{u}_1\}$, then there exists another vector $\vec{u}_2 \in H$, which is not in $\text{Span}\{\vec{u}_1\}$. Now, consider $\text{Span}\{\vec{u}_1, \vec{u}_2\}$. If $H = \text{Span}\{\vec{u}_1, \vec{u}_2\}$, then we are done. Otherwise, pick $\vec{u}_3 \notin \text{Span}\{\vec{u}_1, \vec{u}_2\}$. If we continue this way, the process must stop with $\vec{u}_k$, where $k \leq n$ and thus $H = \text{Span}\{\vec{u}_1, \ldots, \vec{u}_k\}$

In the past, we discussed how in $\mathbb{R}^n$, if $k > n$, then the set $\{\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_k\}$ is linearly dependent.

Since the vectors $\vec{u}_i$ we picked in the proof above are not in the span of previous vectors, they must be linearly independent, and thus we can conclude the following:

Corollary: (Subspaces are spans of linearly independent vectors)

If H is a subspace of $\mathbb{R}^n$, then there exist linearly independent vectors $\{\vec{u}_1, \ldots, \vec{u}_k\}$ of H such that

$$H = span\{\vec{u}_1, \ldots, \vec{u}_k\}$$

Corollary: (Subspaces are spans of linearly independent vectors)

If H is a subspace of $\mathbb{R}^n$, then there exist linearly independent vectors $\{\vec{u}_1, \ldots, \vec{u}_k\}$ of H such that

$$H = span\,\{\vec{u}_1, \ldots, \vec{u}_k\}$$

Example: Is the set $V = \left\{ \begin{bmatrix} a-3b \\ 2a+b \\ a \end{bmatrix}, \ a, b \in \mathbb{R} \right\}$ a subspace of $\mathbb{R}^3$?

Yes, since $\begin{bmatrix} a-3b \\ 2a+b \\ a \end{bmatrix} = a \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + b \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}$, which means

$$V = \mathrm{Span}\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

Hence, from the above Theorem, we can conclude that V is a subspace of $\mathbb{R}^3$. The set $\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix} \right\}$ is a linearly independent spanning set for V.

Note that V can also be written as $V = \mathrm{Span}\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix} \right\}$, where the set $\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix} \right\}$ is a spanning set, which is not linearly independent.

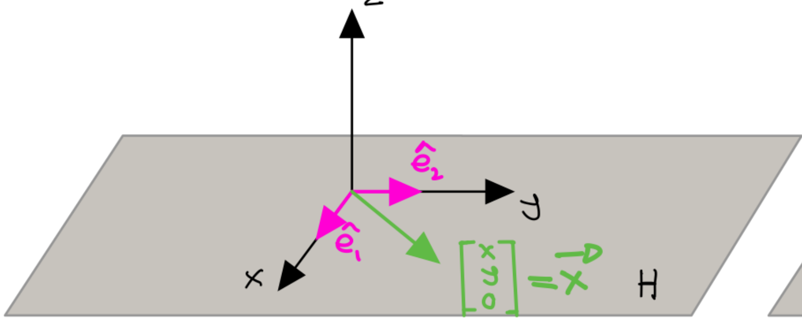
Basis of a Subspace

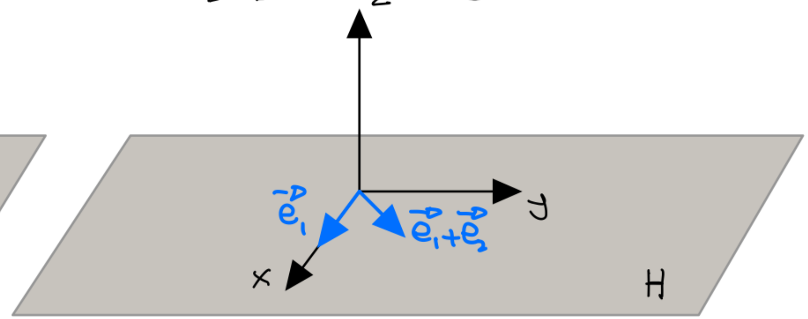
Definition

Let H be a subspace of $\mathbb{R}^n$. Then $\{\vec{u}_1, \ldots, \vec{u}_k\}$ is called a basis for H if the following conditions hold:

1. $\text{span}\{\vec{u}_1, \ldots, \vec{u}_k\} = H$
2. $\{\vec{u}_1, \ldots, \vec{u}_k\}$ is linearly independent.

Example: Consider the xy-plane. i.e., $H = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \;\middle|\; z = 0 \right\}$.





$H = \text{Span}\{\vec{e}_1, \vec{e}_2\}$

$B_1 = \{\vec{e}_1, \vec{e}_2\}$ $\quad \vec{x} = x\vec{e}_1 + y\vec{e}_2$

$H = \text{Span}\{\vec{e}_1, \vec{e}_1 + \vec{e}_2\}$

$B_2 = \{\vec{e}_1, \vec{e}_1 + \vec{e}_2\}$ $\quad \vec{x} = (x-y)\vec{e}_1 + y(\vec{e}_1 + \vec{e}_2)$

Theorem We will be using this theorem, to prove the next theorem. Below is its proof for an interested reader. You can skip it if you wish to.

Suppose $\{\vec{u}_1, \cdots, \vec{u}_r\}$ is a linearly independent set of vectors in $\mathbb{R}^n$, and each $\vec{u}_k$ is contained in span$\{\vec{v}_1, \cdots, \vec{v}_s\}$ Then $s \geq r$.
In words, spanning sets have at least as many vectors as linearly independent sets.

Proof. Since each $\vec{u}_j$ is in span$\{\vec{v}_1, \cdots, \vec{v}_s\}$, there exist scalars a_{ij} such that

$$\vec{u}_j = \sum_{i=1}^{s} a_{ij}\vec{v}_i$$

Suppose for a contradiction that $s < r$. Then the matrix $A = \left[a_{ij}\right]$ has fewer rows, s than columns, r. Then the system $AX = 0$ has a non trivial solution $\vec{d}$, that is there is a $\vec{d} \neq \vec{0}$ such that $A\vec{d} = \vec{0}$. In other words,

$$\sum_{j=1}^{r} a_{ij}d_j = 0,\ i = 1, 2, \cdots, s$$

Therefore,

$$\begin{aligned}\sum_{j=1}^{r} d_j\vec{u}_j &= \sum_{j=1}^{r} d_j \sum_{i=1}^{s} a_{ij}\vec{v}_i \\ &= \sum_{i=1}^{s}\left(\sum_{j=1}^{r} a_{ij}d_j\right)\vec{v}_i = \sum_{i=1}^{s} 0\vec{v}_i = 0\end{aligned}$$

which contradicts the assumption that $\{\vec{u}_1, \cdots, \vec{u}_r\}$ is linearly independent, because not all the d_j are zero. Thus this contradiction indicates that $s \geq r$. ♠

Theorem

Let H be a subspace of $\mathbb{R}^n$ and suppose $\{\vec{u}_1, \ldots, \vec{u}_r\}$ and $\{\vec{v}_1, \ldots, \vec{v}_s\}$ are two bases for H. Then $r = s$.

Note that any spanning set has at least as many vectors as a basis. Also, any basis is also a spanning set. Hence:

$$\left.\begin{array}{l}
\text{(I) } \{\vec{u}_1, \ldots, \vec{u}_r\} \text{ is a basis and } \{\vec{v}_1, \ldots, \vec{v}_s\} \text{ is a spanning set, Therefore } r \leqslant s. \\
\text{(II) } \{\vec{v}_1, \ldots, \vec{v}_s\} \text{ is a basis and } \{\vec{u}_1, \ldots, \vec{u}_r\} \text{ is a spanning set, Therefore } s \leqslant r.
\end{array}\right\} \Longrightarrow$$

$r = s$.

Definition

Let H be a subspace of $\mathbb{R}^n$. Then the dimension of H is the number of a vectors in a basis of H.

Example: What is the dimension of the following subspaces of $\mathbb{R}^3$?

$$H_1 = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \;\middle|\; x = y = z \right\}$$

H_1 is a line through the origin with the direction vector $\vec{v} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $H_1 = \text{Span}\{\vec{v}\}$, hence

$\dim H_1 = 1$.

$$H_2 = \left\{ \vec{u} \in \mathbb{R}^3 \;\middle|\; \vec{u} \cdot \vec{v} = 0,\ \vec{v} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \right\}$$

$\vec{u} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ $\quad \vec{u} \cdot \vec{v} = 2x - y + z = 0$ which is a plane through the origin and

$H_2 = \text{Span}\left\{ \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\}$. Note that any two vectors on the plane that are linearly independent can be used here. Hence $\dim H_2 = 2$.